

Cloud handoff — #485, journalling one fetch is quadratic in the refs it moved

Written: 2026-08-25 · **By:** max (CLI session on Tom's box) · **For:** a cloud Claude Code session on `tom2025b/git-vista`

The measurements are already done, by the session that filed this. **A 500-ref fetch spends 27.6 seconds journalling, on the user's latency, and leaves a 34.7 MB journal.** You do not need to re-derive that. You need to fix it and prove the numbers moved.

```
task_id: gv-485-journal-quadratic
issue: 485
milestone: —          # a #329 successor
repo: tom2025b/git-vista
base: main
branch: fix/485-journal-capture-refs
sign_commits_as:
  name: Claude_Max
  email: 262510778+tom2025b@users.noreply.github.com
sign_artifacts_as: max
adr_number: 0080          # ASSIGNED. Do not pick "the next free" one.
allowed_paths:
  - crates/git-vista-server/src/journal.rs
  - crates/git-vista-server/src/planner/fetch.rs
  - crates/git-vista-server/src/handlers/mod.rs
  - crates/git-vista-core/src/activity.rs
  - docs/adr/
forbidden_paths:
  - design-docs/
  - ci/browser/**
  - crates/git-vista/src/**
  - handoff.md
merge_order: independent — touches no file the stash follow-ups touch.
```

The cause, in one paragraph

`fetch::journal_updates` (`planner/fetch.rs`) loops over the updated refs calling `journal_app_event` once per ref. That reaches `journal::append` (`journal.rs:124`), and because the event arrives with `refs: None` (`handlers/mod.rs:78`), `append` calls `capture_refs` (`journal.rs:96`) on **every iteration**. `capture_refs` does a full ref read of the repository and embeds up to `REFS_PER_EVENT_CAP` (500 — `activity.rs:260`) branches, 500 tags **and** 500 remote-tracking refs into that single line. So a fetch of N refs performs N full ref reads and writes N lines whose size itself grows with N .

The filed measurements, against real `git init` repos and the real `journal::append`:

refs moved	journal bytes	bytes/line	journalling time
1	538 B	537	2.4 ms
94	526.8 KiB	5,736	1,086.7 ms
500	14.1 MiB	28,872	27,606.2 ms

bytes/line going 537 → 28,872 is the quadratic, stated in one column.

And it is on the hot path, not in the background. `journal_updates(...).await` at `fetch.rs:333` completes before the executor returns `FetchStep::Completed`.

What to be careful about

The 24 August work is not the bug and must not be undone. #464/#467/#468 made the journal read bounded, linear and single-copy, and they did exactly what they claimed. This issue is about what each line contains. Do not "optimise" the reader again.

Capture the refs ONCE per fetch, not once per ref is the obvious shape, and it is probably right — but the decision that needs writing down is what a journal line means afterwards. Today every line carries a full ref snapshot, and something downstream may be relying on that being per-line rather than per-batch. `assemble_feed` is where to look. If the answer is "one snapshot for the batch, and the lines reference it", say so in ADR 0080 and say what reads it.

capture_refs embedding 500 remote-tracking refs is its own question. #487 (latent, low) is about push journalling per remote-tracking ref. If your fix makes #487 moot, say so on that issue rather than silently closing it.

What you cannot run

The browser leg does not run in a cloud container — the kernel reports `landlock_abi=-1` and INV-13 gives the server no degraded tier. Installing `bwrap` changes nothing. This task should not need it; if you find yourself wanting it, the change has reached the frontend, which is outside `allowed_paths`.

Two tests in `git-vista-server` flake under parallel execution because they race on the process-global current repository (#438):

`recovery_center::tests::a_stale_claimed_undo_is_refused_and_the_branch_is_left_alone` and `state::tests::selection_flow_carries_mode_and_gates_writes`. Re-run before believing either is your doing.

Acceptance

- bytes/line is flat in the number of refs moved. **Reproduce the filed table's method and put your own row for 500 refs in the PR body.** A claim that it is fixed, without the number beside the old one, is not this issue's deliverable.
- Journalling time for a 500-ref fetch is a small multiple of the 1-ref case, not 11,000×.

3. A test that fails if the per-ref full ref read comes back — and proved able to go red **two different ways** (remove the batching, and weaken it to every-other-ref). One caught verdict is not proof.
 4. **ADR 0080** records what a journal line carries after this change, and what reads it. docs/adr/README.md index updated.
 5. cargo fmt --all, cargo clippy --all-targets -- -D warnings, cargo test --workspace green.
 6. PR body says Closes #485. **Never delete the branch.**
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Signed: max · 2026-08-25T10:25:00-04:00